

Chapter 3

Group Relations

Interlude on Set Theory

We say that two sets have the same **cardinality** if $\exists f : X \hookrightarrow Y$, and we write $|X| \leq |Y|$. This allows us to inter-relate cardinality and function relations via a special theorem.

Theorem 3.0.1 Schröder-Berstein Theorem

If $\exists f : X \hookrightarrow Y$ and $g : Y \hookrightarrow X$ then $|X| = |Y|$.

The Schröder-Berstein theorem provides some immediate results for allowing explicit bijections.

Proposition 3.0.1 Cardinality of $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ to any power

Using Schröder-Berstein we can create explicit bijections $f : \mathbb{Z} \leftrightarrow \mathbb{N}$ given by $n \mapsto$

$$\begin{cases} 2n & \text{if } n \geq 0; \\ -(2n + 1) & \text{if } n < 0. \end{cases}$$

Proposition 3.0.2 $|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}| = \aleph_0$

We create the explicit bijection via $f : \mathbb{N} \times \mathbb{N} \leftrightarrow \mathbb{N}$, $(k, l) \mapsto 2^k(2l + 1) - 1$. It follows then that $\aleph_0^a, \aleph_0^b, \aleph_0^c, \forall a, b, c \in \mathbb{N}$.

Proof: Proof of Prop. 3.1.2

Notice that any natural number can be written as $2^k(m)$ where m is odd, i.e., $m = 2n + 1$. A bijection is automatically created here with $2^k(2l + 1)$, and the additional -1 at the end of the statement is there to shift the entire bijection down one. So instead of $(0, 0) \mapsto 2^0(2(0) + 1) = 1$ we get $(0, 0) \mapsto 2^0(2(0) + 1) - 1 = 0$. $\square$

Proposition 3.0.3 Countable vs uncountable infinities

$|\mathbb{R}| > |\mathbb{N}|$.

Proof: Proof by Cantor

Given by Cantor's diagonal argument, no map $f : \mathbb{N} \rightarrow \mathbb{R}$ be surjective, because:

$$\begin{cases} f(0) &= a_{00}a_{01}a_{02}a_{03}\dots \\ f(1) &= a_{10}a_{11}a_{12}a_{13}\dots \\ f(2) &= a_{20}a_{21}a_{22}a_{23}\dots \end{cases}$$

then let $y = b_0b_1b_2b_3\dots$ where $b_j \neq a_{jj}$ for each j . Now look at the j^{th} digit, $y \neq f(j)$, $\forall j \in \mathbb{N}$, so f can't be surjective. $\square$

Proposition 3.0.4 Cardinality of mappings

$$|\mathbb{R}^{\mathbb{N}}| = |\mathbb{R}|$$

Proof: Proof by Cantor's diagonal used for mappings

We can use the fact that $|(0,1)| = |\mathbb{R}|$ in addition to Cantor's diagonal argument as a decimal expansion— $t = 0.t_1t_2t_3\dots$ —with t_i as an integer between 0 and 9. So now to show $\exists f : (0,1) \leftrightarrow (0,1)^{\mathbb{N}}$, we create some $s_j \in (0,1)^{\mathbb{N}}$ and construct the bijection via:

$$\begin{cases} s_1 &= 0.t_1t_3t_5\dots \\ s_2 &= 0.t_2t_6t_{10}\dots \\ s_3 &= 0.t_4t_{12}t_{20}\dots \end{cases}$$

We say that s_1 is made up of all t_i that are odd; s_2 is made up of all i which are odd such that $i \cdot 2 \nmid i \in s_2$; s_3 is made up of all i which are odd such that $i \cdot 2^2 \nmid i \in s_3$, $i \cdot 2^3 \nmid i \in s_3$ and so on (the general form being $i \cdot 2^n \nmid i = j-1$). Therefore there exists a bijection between $(0,1) \leftrightarrow (0,1)^{\mathbb{N}}$. $\square$

Proposition 3.0.5 There does not exist a largest cardinality

Let $\mathcal{P}(S)$ be the standard powerset. It follows then that $\mathcal{P}(S) = \{0,1\}^S$ and $|\mathcal{P}(S)| > |S|$ for any S .

Proof: Proof of non-existence of largest cardinality

Recall that $\{0,1\}^S = \{\text{all } f \text{ such that } f : S \rightarrow \{0,1\}\}$. There must exist a bijection since $|\mathcal{P}(S)| = 2^{|S|}$ and $|\{0,1\}^S| = 2^{|S|}$. Now to show $|\mathcal{P}(S)| > |S|$ we use proof by contradiction: suppose $\exists \alpha : S \leftrightarrow \mathcal{P}(S)$ and let $A \subset S$ defined by $A := \{s \in S \mid s \neq \alpha(s)\}$, now for $A = \alpha(t)$ for some $t \in A$ we notice that $t \notin \alpha(t) = A$ means that $t \in A$, but to say $t \in \alpha(t) \implies t \notin A$, thus we have a contradiction. $\square$

Another option for this proof, though less formal, can be found by showing that $\mathcal{P}(S) = \{0,1\}^S$ and that subsets of S have f such that $f \mapsto f^{-1}$; let $A \subset S$ then

$$A \mapsto \left(\not\llcorner_A : x \mapsto \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases} \right)$$

Note that if S is finite then $|S| = n$ and $\mathcal{P}(S) = 2^n$. This naturally leads us into our next theorem regarding the infinitude of the cardinality of S in relation to $\mathcal{P}(S)$.

3.1 Quotient groups

Consider $\mathbb{Z}/n$. Under the equivalence relation congruence mod n we can take the sum of the various subsets that mod n partitions $\mathbb{Z}$ into as the law of composition given by $(a + n\mathbb{Z}) + (b + n\mathbb{Z}) = a + b + n\mathbb{Z}$. Notice that we can let T be the set of equivalence classes under mod n , and thus we have a homomorphic map from $\mathbb{Z} \rightarrow T$. Finally, $T = \mathbb{Z}/n$.

Why stop at integers? What if we want to make congruence mod H where H is a subgroup $H \subset G$? Under what cases, given that we *can* make mod H equivalence classes (we will define this in a bit) is the quotient map $G \rightarrow T$ a homomorphism?

3.1.1 Congruence mod H

For any two elements $a, b \in G$ we define $a \sim b \iff ab^{-1} \in H$, this is to say that $a \in bH$. We thus have $a \sim a$ since $e \in H$, $a \sim b \implies ab^{-1} \in H$, and $ba^{-1} = (ab^{-1})^{-1} \in H$, thus $b \sim a$. Finally, $a \sim b, b \sim c$ meaning that ab^{-1} and bc^{-1}

3.2 Symmetry group

3.3 More group relations

3.3.1 Kernels, images, and order

Definition 3.3.1: Kernel

The *kernel* of a group homomorphism, $\varphi : G \rightarrow H$, is given by $\ker(\varphi) = \{a \in G \mid \varphi(a) = e_H\}$.

Proposition 3.3.1

The kernel is a subgroup of G and φ is injective $\iff \ker(\varphi) = \{e_G\}$ ($\varphi(a) = \varphi(b) \iff a^{-1}b \in \ker(\varphi)$).

Definition 3.3.2: Image of group homomorphisms

The image of a group homomorphism $\varphi : G \rightarrow H$ is $\text{Im}(\varphi) = \varphi(G) = \{b \in H \mid \exists a \in G : \varphi(a) = b\}$. This is a subgroup of H and φ is surjective $\iff \text{Im}(\varphi) = H$.

Note:-

If φ is injective, then G is isomorphic to the subgroup $\text{Im}(\varphi) \subset H$. (if the isom. is given by the map $G \rightarrow \text{Im}(\varphi), a \mapsto \varphi(a)$).

3.3.2 Finite symmetric groups